

Infosys Springboard Virtual Internship 7.0 Completion Report

Team Name: Team A

Team Members:

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Internship Duration: 8 weeks

Year: 2026

1. Project Title

Cyber Threat Analytics Platform with Incident Visualization System Group 1

2. Project Objective

The main objective of this project was to design and build a visual-first cybersecurity analytics framework that replaces slow, error-prone manual log review with an interactive Power BI dashboard. Working from raw network-traffic and threat-detection logs, the project first engineered a clean, analysis-ready dataset with risk-indicator fields — Severity_score, Urgency_Flag and Incident_ID — and then used that foundation to build a two-page interactive dashboard. The dashboard lets security analysts and non-technical stakeholders instantly see traffic patterns, protocol usage, and network origins, and — most importantly — see which malicious threats were actually blocked versus missed. This directly supports Security Operations Centres (SOCs), where faster, visual triage of incidents narrows the gap between detection and prevention, and gives the organisation a data-driven basis for improving its security posture.

3. Project Description

3.1 Overview

In the modern digital landscape, cybersecurity is no longer solely about deploying defensive measures to stop attacks; it requires an ongoing, deep understanding of massive volumes of network data in real time. Traditional security operations rely heavily on text-based log analysis, which limits the speed at which threats can be identified and remediated. This project tackles this 'data overload' problem driven by massive log volume, analytical complexity, and the bottleneck of manual review by replacing static log reading with interactive data exploration. Using the Kaggle 'Cybersecurity Threat Detection Logs' dataset (an enterprise-scale set of network-traffic and security events).

The project builds a clean data foundation in Power Query and then a fully interactive Power BI dashboard so that hidden geographic/network origins, time-based attack patterns, and critical network breaches can be spotted at a glance.

3.2 Project Approach

1. **Problem Understanding:** Defining the core challenge of cybersecurity data overload and identifying manual log analysis as the primary operational bottleneck.
2. **Data Collection:** Sourcing the 6-million-row raw CSV dataset ('Cybersecurity Threat Detection Logs') from Kaggle.
3. **Data Understanding:** Analyzing the required dataset dimensions, including temporal (timestamps), geospatial (source and destination IPs), and security metrics (actions and threat labels).

4. **Data Connection:** Establishing the initial connection to the Power BI environment and leveraging the engine's active memory compression.
5. **Data Preprocessing (12-Step ETL):** Executing a rigorous Power Query pipeline across three phases (Scale Optimization, Chronology & Text Format, and Enrichment & Finalization) to clean and optimize the data.
6. **Data Modeling (Close & Load):** Establishing table relationships, configuring data indexing, and preparing Data Analysis Expressions (DAX) measures.
7. **Data Visualization:** Designing the interactive UI elements, including regional maps for geospatial analysis, area charts for time-series trends, and KPI scorecards for threat highlighting.
8. **Share / Publish Report:** Deploying the final executive dashboard to stakeholders, delivering actionable insights and facilitating rapid decision-making.

3.3 Technology Used

- Platform: Microsoft Power BI Desktop
- Data Preparation: Power Query (M language)
- Analytics & Visuals: DAX measures and calculated columns, Power BI report visuals (KPI cards, line/area charts, donut charts, treemap, stacked bar chart, table)
- Dataset: 'Cybersecurity Threat Detection Logs' (Kaggle) cybersecurity_threat_detection_logs table, 1,495,242 network log events spanning January–March 2024, reduced to a clean 13-column feature-enriched table.
- Version Control & Repository Management: GitHub, GitLab
- Collaboration & Communication: Microsoft Teams

3.4 Insights from the Dashboard

The development and deployment of the Advanced Threat Incident Analytics Framework transformed 1.5 million raw network logs into actionable security intelligence. By cross-filtering protocols, severity levels, and network origins, the dashboard revealed several critical insights regarding the organization's current security posture:

3.4.1 Severe Perimeter Defense Vulnerabilities

One of the most critical discoveries was the identification of a **49.89% Critical Breach Rate**. While the firewall successfully blocks a high volume of standard, low-severity noise, the dashboard revealed that nearly half of all highly malicious, targeted traffic successfully bypassed perimeter defenses and was flagged as "Allowed." This indicates that existing automated firewall rules are insufficient for advanced threat detection and require immediate recalibration.

3.4.2 Protocol-Specific Attack Vectors and Exfiltration Risks

The Protocol Behavior Analysis isolated specific channels that attackers are exploiting. The data showed that **TCP** accounted for an overwhelming 9.3 GB of data transferred. In a real-world SOC environment, this disproportionate volume strongly indicates potential data exfiltration. Furthermore, the threat distribution charts highlighted a concentrated volume of malicious payloads targeting **FTP** and **SSH** protocols, suggesting active attempts at credential harvesting and lateral movement by external threat actors.

3.4.3 Disproportionate Threat Origins

The Source Network Analysis provided deterministic intelligence on attacker origins. Out of the millions of logged events, the dashboard isolated just **82 distinct external networks** responsible for all external malicious incidents. More specifically, the analytics pinpointed a single aggressor—**Subnet 88**—as the source of nearly 8,900 distinct attacks. This insight transitions the SOC from a reactive posture to a proactive one, providing the exact data needed to blacklist high-risk subnets at the perimeter router.

3.4.4 Isolated Internal Target Vulnerabilities

Beyond tracking external threats, the destination network mapping revealed the internal “blast radius” of these attacks. The data proved that threat actors are not spraying the network randomly; rather, they are focusing heavy fire on specific internal nodes. The dashboard identified machine **192.168.1.217** as the most highly targeted internal IP. This insight allows security engineers to immediately isolate vulnerable internal machines for forensic scanning before lateral breaches can occur.

3.5 Real-World Impact for Media and Public Communication

Beyond the SOC, dashboards like this matter because they translate technical log data into a form that non-technical stakeholders—executives, compliance teams, and communications/PR staff—who may need to brief the media or the public after an incident can understand at a glance.

A single, clear figure such as ‘nearly half of malicious traffic was not blocked’ is far more useful in a leadership briefing or a public disclosure statement than a table of raw log rows, and the underlying KPI/DAX logic means that figure can be trusted and quickly re-verified. By making security posture visually explicit, the framework helps organisations communicate incidents accurately, respond faster to stakeholder and public queries, and build the kind of transparent, data-backed reporting that regulators and the public increasingly expect after a breach.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understand project scope; identify the problem of security-log data overload; collect and evaluate the raw dataset.	Completed – Finalized project title and problem statement; collected the 6-million-row 'Cybersecurity Threat Detection Logs' dataset and completed the data understanding phase.
Week 2	Connect to the raw log source in Power BI and execute a comprehensive data cleaning and transformation pipeline.	Completed – Successfully applied the rigorous 12-step ETL process in Power Query (including scale optimization, chronology formatting, and feature enrichment).
Week 3	Design and develop the initial data visualization framework and establish the Power BI data model and DAX measures.	Completed – Established the core Power BI data model and built the foundational DAX measures for the dashboard.
Week 4	Build the first two interactive dashboard pages of the framework.	Completed – Built the first two interactive pages of the dashboard (Network Traffic & Protocol Overview, and Threat Intelligence & Incident Severity Analysis).
Week 5	Expand the dashboard with protocol-level and severity-level tactical analytics.	Completed – Built Dashboard 3 (Protocol & Action Behavior Analysis) and Dashboard 4 (Severity & Incident Urgency Analysis).
Week 6	Extend the framework with network origin and destination mapping; finalize the three-page tactical expansion.	Completed – Built and finalized Dashboard 5 (Source & Destination Network Analysis), completing the five-page framework.
Week 7	Consolidate the framework into a single executive-facing risk summary.	Completed – Built Dashboard 6 (Executive Risk & Detection Overview), combining a composite risk gauge, headline KPIs, and a goal-tracked detection trend.
Week 8	Test and validate the relationship between transfer volume and malicious activity; finalize the framework.	Completed – Built and finalized Dashboard 7 (High-Volume Malicious Traffic Analysis), completing the end-to-end 7-page interactive Security Operations Center (SOC) framework.

5a. Key Milestones

Milestone	Description	Date Achieved
Milestone 1: Data Collection & Preprocessing (ETL)	Defined the problem statement, acquired the 6-million-row raw cybersecurity dataset, and executed a rigorous 12-step Power Query ETL pipeline to clean, format, and enrich the data.	02-07-2026
Milestone 2: Initial Visualization	Engineered core DAX measures and developed the first two interactive dashboard pages.	16-07-2026
Milestone 3: Tactical Analytics Expansion	Designed and deployed three additional analytical pages covering protocol behavior, severity/urgency analysis, and source & destination network mapping.	06-08-2026
Milestone 4: Executive Risk & Deployment	Delivered the final two dashboard pages — an executive risk summary and a high-volume traffic analysis — completing the end-to-end 7-page SOC framework.	13-08-2026

During this phase of the project, our team successfully transitioned from raw data acquisition to a fully functional, seven-page visualization framework across four distinct milestones:

- **Milestone 1:** Focused on problem definition, data acquisition, and preprocessing — sourcing and analyzing the raw 6-million-row security dataset, then executing a strict 12-step ETL pipeline in Power Query to clean, format, and enrich the logs.
- **Milestone 2:** Transitioned to visual analytics, involving the creation of core DAX measures and the development of the first two baseline dashboard pages (Network Traffic & Protocol Overview, and Threat Intelligence & Incident Severity Analysis).
- **Milestone 3:** Expanded the dashboard with advanced tactical analytics, deploying the next three interactive pages (Protocol & Action Behavior Analysis; Severity & Incident Urgency Analysis; and Source & Destination Network Analysis).
- **Milestone 4:** Concluded with an executive risk summary and a high-volume traffic analysis, deploying the final two pages (Executive Risk & Detection Overview, and High-Volume Malicious Traffic Analysis) to complete the end-to-end 7-page SOC framework.

5b. Project Execution Details

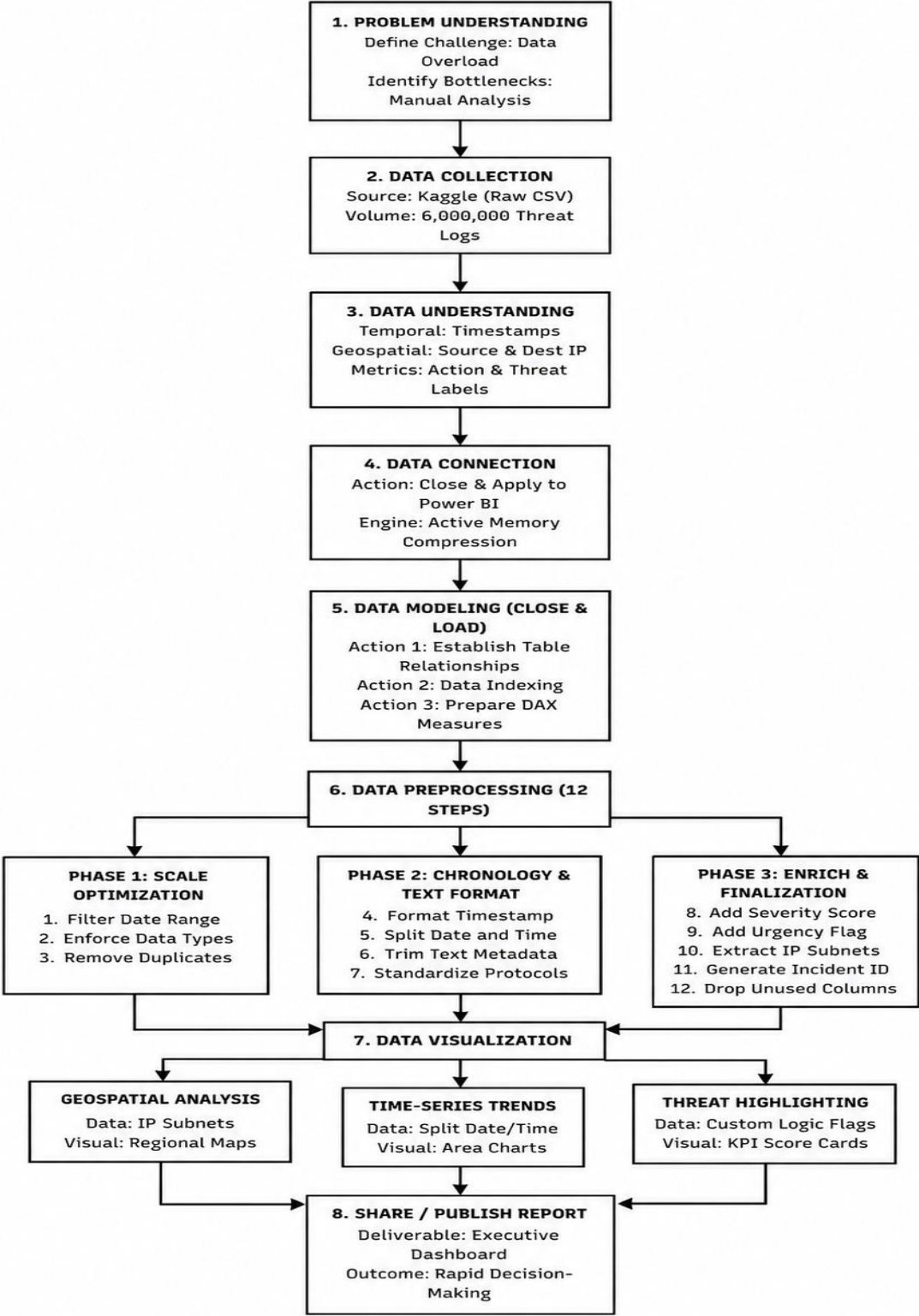


Fig 5.1 Workflow Diagram

1. Problem Understanding

- Define Challenge: The core challenge driving this project is "Data Overload" within security operations centers. Modern enterprise systems generate millions of raw threat and incident logs daily, causing severe data fatigue for analysts.
- Identify Bottlenecks: The primary operational bottleneck is relying on manual raw data analysis. This process is incredibly slow, highly prone to human error, and creates dangerous gaps in incident response times.

2. Data Collection

- Source: The data was sourced from Kaggle as a raw CSV file titled 'Cybersecurity Threat Detection Logs'.
- Volume: The dataset contains exactly 6,000,000 rows of synthetic network traffic and security events, which perfectly simulates a true corporate data overload environment.

3. Data Understanding

- Temporal: Timestamps capture the exact date and time of the network events, serving as the foundation for time-series trendlines.
- Geospatial: Source and Destination IPs track the origin and target network addresses to enable geospatial mapping and cluster identification.
- Metrics: The data relies on 'Action' metrics (indicating if traffic was allowed or blocked) and 'Threat Labels' (classifying intent as benign, suspicious, or malicious).

Field Name	Description & Visualization Map
Timestamp	Captures the exact date/time of the network event; serves as the foundation for time-series trendlines.
Source & Dest. IP	Tracks the origin and target network addresses; enables geospatial mapping and cluster identification.
Protocol	Denotes the network protocol utilized (e.g., TCP, UDP, HTTP); allows for traffic categorization.
Log Type	Identifies the system that recorded the event (e.g., Firewall, IDS, Application).
Action	Indicates the system's response to the event (e.g., 'allowed' or 'blocked').
Threat Label	Classifies the event's intent as 'benign', 'suspicious', or 'malicious'.

4. Data Connection

- Action: The dataset was ingested into the Power BI environment using the "Close & Apply" protocol.
- Engine: This step leverages Power BI's active memory compression engine to ensure the platform can handle the massive dataset without system lag.

5. Data Modeling (Close & Load)

- Actions 1, 2, & 3: This phase involves establishing table relationships, configuring data indexing, and preparing the foundational Data Analysis Expressions (DAX) measures required for advanced analytics.

6. Data Preprocessing (12-Step ETL)

To handle the massive scale, a rigorous 12-step data preparation pipeline was implemented in the Power Query Editor, broken into three specific phases:

- Phase 1: Scale Optimization
 - 1. Filter Date Range: Controlled the data volume to ensure optimal dashboard performance during the development phase.
 - 2. Enforce Data Types: Explicitly defined schemas (e.g., Text vs. Whole Number) to leverage Power BI's compression engine.
 - 3. Remove Duplicates: Cleansed identical network echoes to ensure pristine mathematical accuracy for incident counting.
- Phase 2: Chronology & Text Format
 - 4. Format Timestamp: Converted raw text fields into strictly recognized Date/Time data types.
 - 5. Split Date and Time: Separated the time variables into distinct columns to enable drill-down hourly trendline analysis.
 - 6. Trim Text Metadata: Removed hidden formatting spaces from text columns to ensure accurate dashboard filtering.

- 7. Standardize Protocols: Forced uppercase formatting on network protocols to clean up visual legends and group data accurately.
- Phase 3: Enrich & Finalization
 - 8. Add Severity Score: Created a custom numeric scale (1-3) based on threat classifications to enable KPI tracking.
 - 9. Add Urgency Flag: Built custom conditional logic to instantly highlight "Critical Breaches" (defined as Malicious threats where the Action was Allowed).
 - 10. Extract IP Subnets: Extracted the text before the first delimiter in the Source IP to group IPs by top-level network class, preventing map visual crashing.
 - 11. Generate Incident ID: Created a unique integer primary key to accelerate DAX counting metrics.
 - 12. Drop Unused Columns: Removed high-cardinality metadata (e.g., User Agent, Request Path) to finalize a lightweight, high-performance data model.

Building on the cleaned and structured dataset produced in the data-preparation phase, this dashboard was developed in Power BI across two report pages, each built around seven visuals. Page 1 profiles overall network traffic and protocol usage, while Page 2 focuses specifically on threat detection and incident severity. The dashboard draws on the `cybersecurity_threat_detection_logs` table (1,495,242 network log events spanning January to March 2024), with each row representing a single network event – a source IP communicating with a destination IP over a given protocol, that was either allowed or blocked, and labelled benign, suspicious, or malicious. The sections below describe each visual in detail, along with the DAX measures and calculated columns used to build them.

Dashboard 1 – Network Traffic & Protocol Overview

This page answers one core question: what does normal network activity look like, before the analysis turns to threats specifically? It combines three summary KPI cards with four supporting visuals covering daily traffic volume, protocol usage, log-source behaviour, and data-volume distribution.

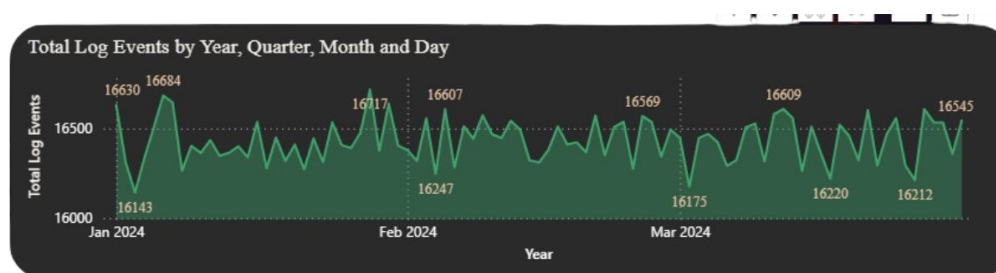
Visuals 1–3: KPI Cards — Total Log Events, % Traffic Blocked, Total Data Transferred



These three cards set the overall scale for the page. They show that the dataset contains approximately 1.5 million logged events, that close to half of all traffic (50%) was blocked, and the total volume of data transferred across the network during the quarter. These figures act as reference points for interpreting the percentages and proportions shown in the remaining visuals on the page.

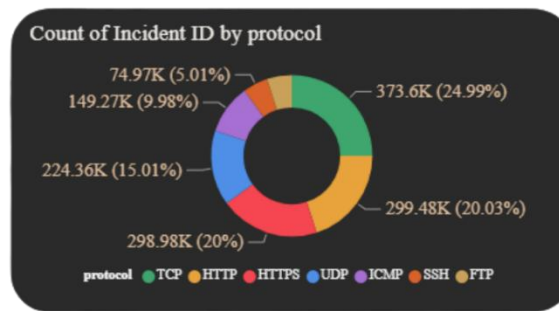


Visual 4: Daily Traffic Trend (Area/Line Chart)



This visual tracks daily event volume across the full quarter, split into allowed versus blocked traffic using Date on the axis and the action field as the legend. The two lines move together fairly consistently from day to day, with no unusual spikes or drop-offs. This indicates that the network was under a steady, predictable load throughout the period rather than being disrupted by a single major incident. No new DAX was required for this visual — it uses the Date column together with the Total Log Events measure, broken down by action.

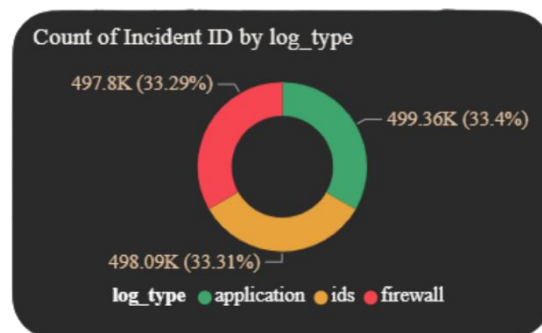
Visual 5: Traffic by Protocol (Donut Chart)



This visual breaks total traffic down by protocol, covering all seven protocols present in the dataset — TCP, HTTP, HTTPS, UDP, ICMP, SSH, and FTP. TCP and the two web protocols (HTTP and HTTPS) account for the largest share of traffic, which is consistent with what would be expected on a typical enterprise network. This breakdown also sets up Visual 7, where the same seven protocols are further grouped by security posture.

```
Traffic Volume Category =
SWITCH(
  TRUE(),
  'cybersecurity_threat_detection_logs'[bytes_transferred] <= 12500, "Low (≤12.5K)",
  'cybersecurity_threat_detection_logs'[bytes_transferred] <= 25000, "Medium (12.5K-25K)",
  'cybersecurity_threat_detection_logs'[bytes_transferred] <= 37500, "High (25K-37.5K)",
  "Very High (>37.5K)"
)
```

Visual 6: Log Volume by Log Type, split by Action



This visual compares three log sources — application logs, IDS logs, and firewall logs — by how much of their traffic was allowed versus blocked. It functions as a sanity check on the network's layered defenses: firewall and IDS logs would be expected to show meaningfully more blocking activity than application logs, since filtering and intrusion detection are specifically their function.

Visual 7: Traffic Volume Category by Protocol Security Type (Treemap)



This visual required two new calculated columns. The first, Protocol Security Type, groups the seven protocols into four broader security categories — Encrypted, Unencrypted, Raw/Transport, and Control/Diagnostic. The second, Traffic Volume Category, buckets the bytes_transferred field into four bands (Low, Medium, High, and Very High) based on the actual quartiles of the data, so that the bands are evenly distributed rather than arbitrary cut-offs. Together, the treemap shows where the heaviest data transfers sit across these groups, revealing whether large transfers are predominantly happening over secure or insecure channels.

```
Protocol Security Type =
SWITCH(
    'cybersecurity_threat_detection_logs'[protocol],
    "HTTPS", "Encrypted",
    "SSH", "Encrypted",
    "HTTP", "Unencrypted",
    "FTP", "Unencrypted",
    "TCP", "Raw/Transport",
    "UDP", "Raw/Transport",
    "ICMP", "Control/Diagnostic",
    "Other"
)

Traffic Volume Category =
SWITCH(
    TRUE(),
    'cybersecurity_threat_detection_logs'[bytes_transferred] <= 12500, "Low
(≤12.5K)",
    'cybersecurity_threat_detection_logs'[bytes_transferred] <= 25000, "Medium
(12.5K–25K)",
    'cybersecurity_threat_detection_logs'[bytes_transferred] <= 37500, "High
(25K–37.5K)",
    "Very High (>37.5K)"
)
```

Dashboard 2 – Threat Intelligence & Incident Severity Analysis

Where Page 1 establishes what normal traffic looks like, Page 2 is the core security story of the dashboard: of all that traffic, what was actually dangerous, and was it caught? This page combines three KPI cards with four supporting visuals covering response status, traffic origin, high-risk source IPs, and the trend of threat activity over time.

Visuals 1–3: KPI Cards — Total Malicious Incidents, Critical Breaches, Malicious Allowed Rate %



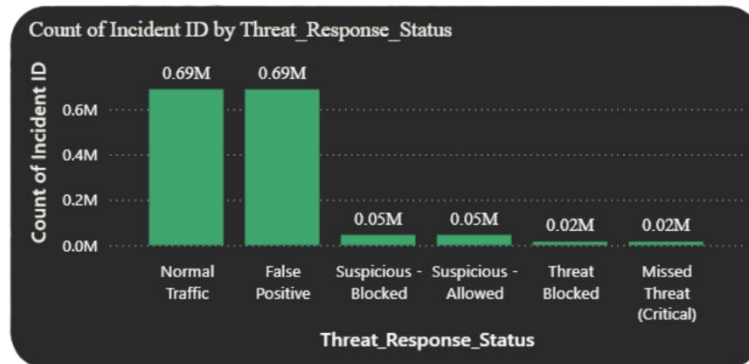
These cards show that approximately 30,000 events in the dataset were labelled malicious. Of those, roughly 15,000 — about 49.89% — were still marked "allowed" rather than blocked. This is the single most important statistic on the dashboard: it means close to half of all malicious traffic passed through the network's defenses undetected or unstopped.

```
Total Malicious Incidents =
CALCULATE(COUNTROWS('cybersecurity_threat_detection_logs'),
'cybersecurity_threat_detection_logs'[threat_label] = "malicious")

Critical Breaches (Missed Threats) =
CALCULATE(
    COUNTROWS('cybersecurity_threat_detection_logs'),
    'cybersecurity_threat_detection_logs'[Threat Response Status] = "Missed Threat
(Critical)"
)

Malicious Allowed Rate % =
DIVIDE(
    CALCULATE(COUNTROWS('cybersecurity_threat_detection_logs'),
'cybersecurity_threat_detection_logs'[threat_label] = "malicious",
'cybersecurity_threat_detection_logs'[action] = "allowed"),
    [Total Malicious Incidents]
)
```

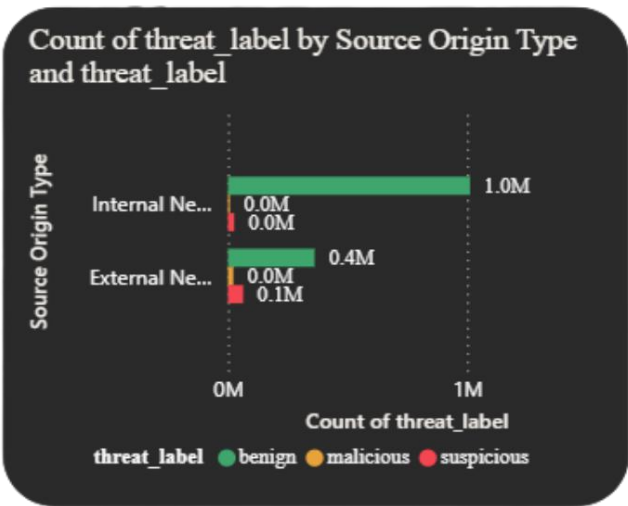
Visual 4: Incident Count by Threat Response Status (Bar Chart)



This is the core chart of the entire dashboard. A calculated column, Threat Response Status, combines the threat label and the action taken into a single, readable status: Missed Threat (Critical), Threat Blocked, Suspicious – Allowed, Suspicious – Blocked, False Positive, or Normal Traffic. The "Missed Threat (Critical)" category is deliberately highlighted in red — even though it is a small bar relative to total traffic, it represents genuine malicious activity that made it past the network's defenses, making it the bar most likely to concern a security team.

```
Threat Response Status =
SWITCH(
    TRUE(),
    'cybersecurity_threat_detection_logs'[threat_label] = "malicious" &&
    'cybersecurity_threat_detection_logs'[action] = "allowed", "Missed Threat (Critical)",
    'cybersecurity_threat_detection_logs'[threat_label] = "malicious" &&
    'cybersecurity_threat_detection_logs'[action] = "blocked", "Threat Blocked",
    'cybersecurity_threat_detection_logs'[threat_label] = "suspicious" &&
    'cybersecurity_threat_detection_logs'[action] = "allowed", "Suspicious - Allowed",
    'cybersecurity_threat_detection_logs'[threat_label] = "suspicious" &&
    'cybersecurity_threat_detection_logs'[action] = "blocked", "Suspicious - Blocked",
    'cybersecurity_threat_detection_logs'[threat_label] = "benign" &&
    'cybersecurity_threat_detection_logs'[action] = "blocked", "False Positive",
    "Normal Traffic"
)
```

Visual 5: threat_label by Source Origin Type (Stacked/Clustered Bar Chart)



This visual is built on a calculated column, Source Origin Type, which classifies each event's source IP as either Internal Network (addresses in the 192.168.x.x range) or External Network (everything else). The chart then splits benign, suspicious, and malicious traffic across these two origins, showing whether threats are predominantly coming from outside attackers or from within the internal network — an important distinction for how a security team would respond to each case.

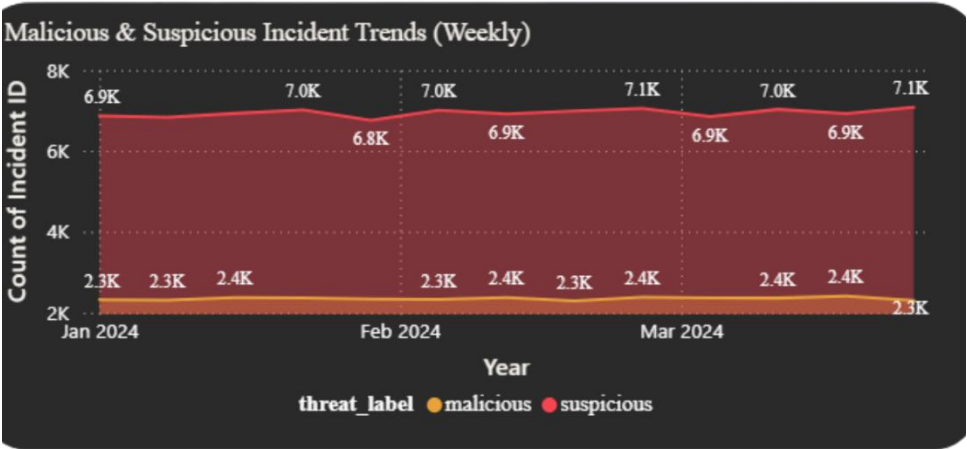
```
Source Origin Type =
IF(
    'cybersecurity_threat_detection_logs'[IP_Network_Origin] = "192",
    "Internal Network",
    "External Network"
)
```

Visual 6: Top 10 Source IPs by Malicious Incident Count (Table)

source_ip	Count of Incident ID
49.138.115.36	271
30.175.227.13	253
8	
221.93.148.50	252
236.1.154.83	251
246.196.87.27	251
150.110.230.1	250
50	
217.89.155.68	247
25.01.54.172	247
Total	2512

This table is filtered to malicious traffic only, then ranked using a Top N filter on source IP by the number of malicious incidents each one triggered. No new DAX was required — this uses a Top N visual-level filter over the existing threat_label and source_ip fields. The result is a directly actionable list: if this dashboard were handed to a SOC analyst, this table is where they would begin their investigation.

Visual 7: Weekly Trend of Malicious & Suspicious Incidents (Line Chart)



This visual uses week number rather than daily granularity, since the dataset's raw time field carried no meaningful hour-of-day detail, which made daily counts too noisy to read a clear trend from. It is filtered to malicious and suspicious traffic only, excluding benign events, so that the trend of threat activity — rising, falling, or flat — can be read clearly across the quarter.

```
Week Number = WEEKNUM('cybersecurity_threat_detection_logs'[Date])
```

Dashboard 3 – Protocol & Action Behavior Analysis

Where Page 1 introduced protocol usage at a glance, this page asks a narrower question: how does each individual protocol behave, both in terms of the volume it carries and how much of its traffic is stopped versus let through? It combines two summary cards with five supporting visuals covering data volume by protocol, allow/block outcomes, threat-label mix, and protocol security classification, alongside protocol, action, and date slicers for drilling into any single view.

KPI Cards: Total Allowed Traffic, Avg Bytes Transferred, and First Protocol (Min)

This page carries three summary cards, each read individually below.



Card 1: Total Allowed Traffic

This card counts the distinct incidents marked “allowed” across the whole dataset: 747,777, which is roughly half of all 1,495,242 events — consistent with the 49.99% blocked figure reported by the % Traffic Blocked card on Page 1. The measure uses DISTINCTCOUNT on Incident ID rather than a plain COUNTROWS; since Incident ID is a unique index generated for every row, the two would return the same number here, but DISTINCTCOUNT is the safer choice if the table were ever joined to a source where Incident ID could repeat.

```
Total Allowed Traffic =  
CALCULATE(  
    DISTINCTCOUNT('cybersecurity_threat_detection_logs'[Incident ID]),  
    'cybersecurity_threat_detection_logs'[action] = "allowed"  
)
```

Card 2: Avg Bytes Transferred

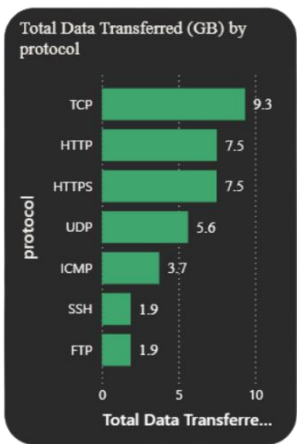
This card shows the mean payload size per event across all protocols and actions combined: approximately 25,028 bytes. It acts as the baseline against which the Traffic Volume Category bands from Page 1 (Low $\leq 12.5K$, Medium 12.5K–25K, High 25K–37.5K, Very High $> 37.5K$) were set, since those cut-offs were built from the quartiles of this same bytes_transferred field.

```
Avg Bytes Transferred =  
AVERAGE('cybersecurity_threat_detection_logs'[bytes_transferred])
```

Card 3: Count of Protocol

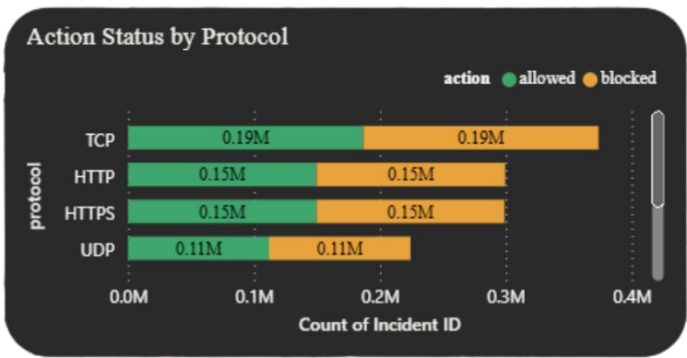
This card is not a DAX measure — it is bound directly to the protocol column using Power BI's built-in Minimum aggregation. With no filter applied it simply returns “FTP,” the alphabetically first protocol name in the table (FTP, HTTP, HTTPS, ICMP, SSH, TCP, UDP). In practice it behaves less as an analytical statistic and more as a live read-out of whichever protocol the page's protocol slicer currently isolates, giving the page a small textual anchor for whatever selection is active.

Visual 1: Total Data Transferred (GB) by Protocol (Bar Chart)



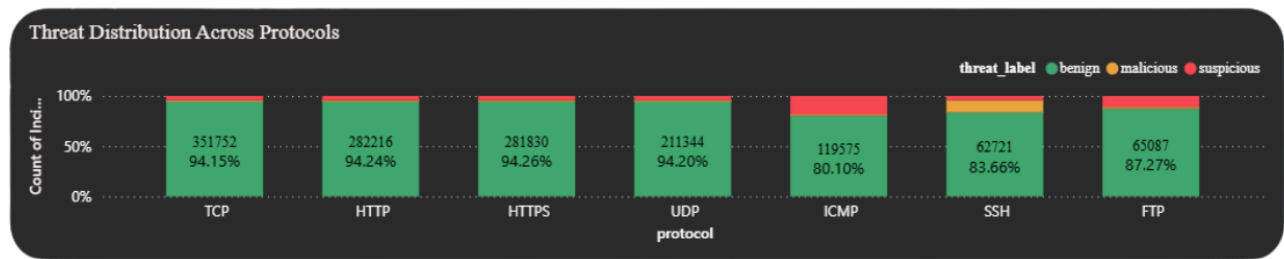
This chart reuses the Total Data Transferred (GB) measure from Page 1, this time broken out by protocol. TCP carries the largest share (about 9.3 GB), followed closely by HTTP and HTTPS (roughly 7.5 GB each), then UDP (5.6 GB), ICMP (3.7 GB), and SSH and FTP tied at the low end (about 1.9 GB each) — a distribution that tracks each protocol's overall event count rather than revealing any single protocol as disproportionately data-heavy.

Visual 2: Action Status by Protocol (Bar Chart)



This chart splits each protocol's event count into allowed versus blocked using the action field as the series. Every protocol lands close to an even 50/50 split (for example, TCP: 186,882 allowed vs. 186,717 blocked), confirming that the roughly 50% block rate seen on Page 1 is not being driven by any one protocol — it holds uniformly across the network's traffic mix. No new DAX was required; this visual reuses the Total Log Events measure filtered by the existing protocol and action fields.

Visual 3: Threat Distribution Across Protocols (100% Stacked Column Chart)



This chart normalizes each protocol's traffic to 100% and splits it by threat_label. Most protocols cluster tightly around 94% benign, 4.3% suspicious, and 1.5% malicious — but SSH is a clear outlier at 12.05% malicious, roughly eight times the rate of any other protocol, while ICMP stands out on the suspicious side at 18.41%. This is the key finding on this page: a security team scanning Page 1's protocol donut alone would have no reason to single out SSH, but this normalized view makes its disproportionate malicious share immediately visible.

Visual 4: Protocol Security Classification (Pivot Table)

Protocol Security Classification					
protocol	Control/Diagnostic	Encrypted	Raw/Transport	Unencrypted	Total
FTP				74578	74578
HTTP				299480	299480
HTTPS		298979			298979
Total	149274	373949	597961	374058	1495242

This matrix cross-tabulates protocol against the Protocol Security Type calculated column introduced on Page 1 (Encrypted, Unencrypted, Raw/Transport, Control/Diagnostic), giving exact event counts for each protocol within its classification bucket. It functions mainly as a validation view — confirming, for instance, that all ICMP traffic (149,274 events) falls under Control/Diagnostic and all HTTPS and SSH traffic falls under Encrypted — rather than surfacing a new pattern of its own.

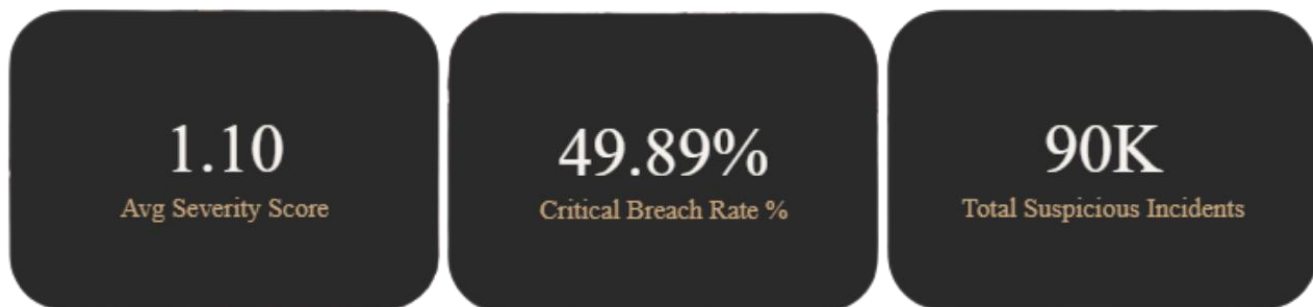
Together with the protocol, action, and date slicers on this page, Dashboard 3 turns the protocol-level summaries from Page 1 into a drill-down view, and its central finding — SSH's outsized malicious rate — gives a security team a specific protocol to prioritize for further investigation alongside the source-IP list surfaced on Page 2.

Dashboard 4 – Severity & Incident Urgency Analysis

Building on the detection-gap finding from Page 2, this page asks a related question: of all logged events, how severe were they, and how urgently would each require triage? It combines three summary cards with four supporting visuals covering the severity distribution, severity by protocol, and the relationship between incident urgency and the action actually taken, alongside severity, urgency, and log-type slicers.

KPI Cards: Total Suspicious Incidents, Critical Breach Rate %, Avg Severity Score

This page carries three summary cards, each read individually below.



Card 1: Total Suspicious Incidents

This card counts every event labelled “suspicious” in threat_label: 90,244 events, or about 6.0% of all 1,495,242 logged events. It sizes the “gray zone” of traffic that sits between benign and malicious — flagged as worth a second look, but not confirmed as an attack — which Page 2's KPI cards did not break out on their own (Page 2 covers Total Malicious Incidents specifically). Taken alongside the 30,473 malicious events, suspicious traffic is roughly three times larger in volume, meaning any SOC workload sized purely around confirmed malicious incidents would be undercounting the actual review queue by a wide margin.

```
Total Suspicious Incidents =
CALCULATE(COUNTROWS('cybersecurity_threat_detection_logs'),
'cybersecurity_threat_detection_logs'[threat_label] = "suspicious")
```

Card 2: Critical Breach Rate %

This card divides Total Critical Urgency Incidents (events where Incident_Urgency = “Critical Breach”) by Total Malicious Incidents, giving $15,203 \div 30,473 = 49.89\%$. That figure is worth flagging explicitly: it is mathematically identical to the Malicious Allowed Rate % measure defined on Page 2, which divides malicious-and-allowed events by total malicious events. The two measures reach the same 49.89% by two

independent routes — one filtering directly on threat_label and action, the other going through the Incident_Urgency column engineered in Power Query — which cross-validates that the Incident_Urgency flag (“malicious AND allowed”, by its Power Query definition) lines up exactly with the DAX-side logic built later. Critical Breach Rate % is the more reusable of the two for this page, since it stays consistent with the Incident_Urgency-based Card 3 on Page 4's urgency chart.

```
Total Critical Urgency Incidents =
CALCULATE(
    COUNTROWS('cybersecurity_threat_detection_logs'),
    'cybersecurity_threat_detection_logs'[Incident_Urgency] = "Critical Breach"
)

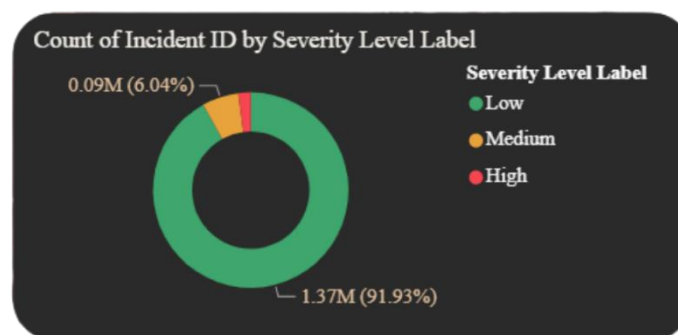
Critical Breach Rate % =
DIVIDE([Total Critical Urgency Incidents], [Total Malicious Incidents])
```

Card 3: Avg Severity Score

This card averages the Severity_Score column (1 = Low, 2 = Medium, 3 = High) across every event, giving 1.10. The low average is expected rather than a sign of a problem: Severity_Score mirrors threat_label directly (1,374,525 Low events, 90,244 Medium, 30,473 High), and since roughly 91.9% of all traffic is benign, a score sitting just above the 1.0 floor is exactly what the underlying label distribution implies.

```
Avg Severity Score =
AVERAGE('cybersecurity_threat_detection_logs'[Severity_Score])
```

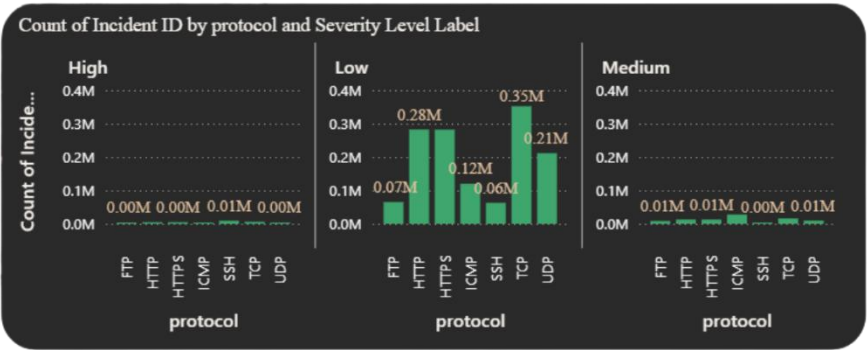
Visual 1: Incident Count by Severity Level (Donut Chart)



This chart visualizes the Severity Level Label calculated column, which maps the Power Query-engineered Severity_score (1–3) onto readable labels. The split is Low: 1,374,525 events (91.9%), Medium: 90,244 (6.0%), and High: 30,473 (2.0%) — the same 91.9/6.0/2.0 split as benign/suspicious/malicious, since Severity_score was derived directly from threat_label back in Milestone 1.

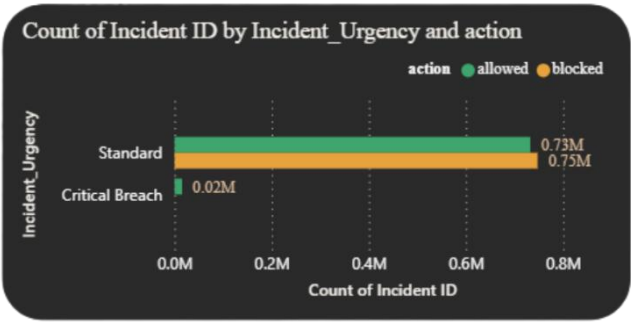
```
Severity Level Label =  
SWITCH(  
  'cybersecurity_threat_detection_logs'[Severity_Score],  
  1, "Low",  
  2, "Medium",  
  3, "High",  
  "Unknown"  
)
```

Visual 2: Incident Count by Severity Level and Protocol (Clustered Column Chart)



This chart groups the same severity split by protocol, and it reinforces the Page 3 finding from a different angle: SSH carries 9,033 High-severity events out of roughly 74,970 total SSH events — about 12% — versus roughly 1.5% for most other protocols. Whether the lens is threat_label (Page 3) or Severity_score (here), SSH consistently surfaces as the protocol with the heaviest concentration of dangerous traffic.

Visual 3: Incident Urgency by Action (Clustered Bar Chart)



This chart compares the two Incident_Urgency values — Critical Breach and Standard — against the action taken. Because Critical Breach is defined in Power Query as “malicious AND allowed,” every one of its 15,203 events is, by construction, allowed (0 are blocked); Standard events split almost evenly, with 732,574 allowed and 747,465 blocked. The chart makes that definitional link visible rather than discovering it: it exists to remind a viewer that the Critical Breach flag is not a probability, it is a certainty by design.

Dashboard 5 – Source & Destination Network Analysis

The final page shifts the framework's focus from what happened to where it came from and where it went. It combines four summary cards with five supporting visuals mapping the internal/external origin split, per-network-origin activity, destination concentration, and threat mix by origin, building directly on the source_ip and dest_ip fields flagged back in Milestone 1 as enabling “geospatial mapping and cluster identification.”

KPI Cards: Distinct Source IP Count, Distinct External Networks, External and Internal Malicious Incidents



This page carries four summary cards, each read individually below.

Card 1: Distinct Source IP Count

This card counts the distinct values in source_ip: 354 unique addresses generated the dataset's 1,495,242 events, meaning each address logged an average of roughly 4,200 events over the quarter. It sets the scale for every other visual on this page — the network-origin bar chart and the Source Origin Type split are both ways of grouping this same pool of 354 addresses.

```
Distinct Source IP Count =  
DISTINCTCOUNT('cybersecurity_threat_detection_logs'[source_ip])
```

Card 2: Distinct External Networks

This card counts the distinct IP_Network-Origin values restricted to rows where Source Origin Type = “External Network”: 82 distinct external network origins. Since the Internal Network segment sits entirely within the single 192.168.x.x block, the dataset's full IP_Network-Origin cardinality is effectively 83 values — the one internal origin plus these 82 external ones — which is the figure the network-origin bar chart draws from.

```
Distinct External Networks =
CALCULATE(
    DISTINCTCOUNT('cybersecurity_threat_detection_logs'[IP_Network_Origin]),
    'cybersecurity_threat_detection_logs'[Source Origin Type] = "External Network"
)
```

Card 3: External Malicious Incidents

This card filters Total Malicious Incidents down to rows where Source Origin Type = “External Network”: 22,538 events. Set against External Network's 30.0% share of total traffic, this is the figure that drives the page's central finding — external sources are responsible for roughly 74% of all malicious activity (22,538 of 30,473) despite carrying less than a third of total events.

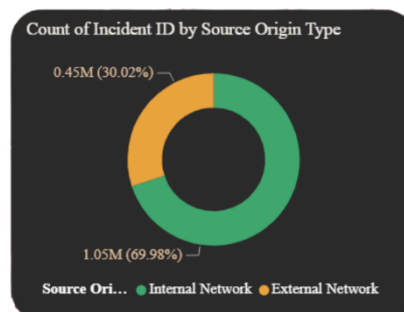
```
External Malicious Incidents =
CALCULATE(
    COUNTROWS('cybersecurity_threat_detection_logs'),
    'cybersecurity_threat_detection_logs'[threat_label] = "malicious",
    'cybersecurity_threat_detection_logs'[Source Origin Type] = "External Network"
)
```

Card 4: Internal Malicious Incidents

This card is the mirror image of Card 3, filtering Total Malicious Incidents to Source Origin Type = “Internal Network” instead: 7,935 events. The two cards are complementary by construction — $22,538 + 7,935 = 30,473$, exactly matching Total Malicious Incidents from Page 2 — which confirms that Source Origin Type cleanly partitions every malicious event into one bucket or the other with no overlap and no gap.

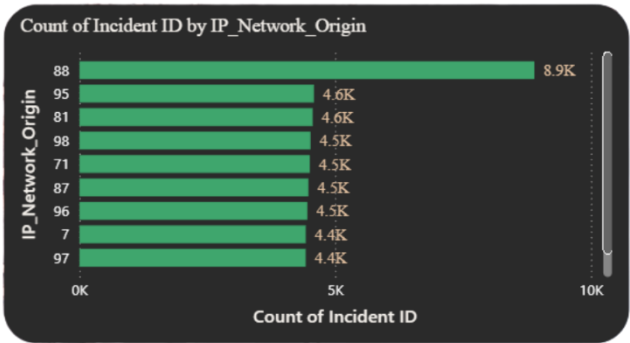
```
Internal Malicious Incidents =
CALCULATE(
    COUNTROWS('cybersecurity_threat_detection_logs'),
    'cybersecurity_threat_detection_logs'[threat_label] = "malicious",
    'cybersecurity_threat_detection_logs'[Source Origin Type] = "Internal Network"
)
```

Visual 1: Traffic Split by Source Origin Type (Donut Chart)



This chart shows that Internal Network traffic accounts for 1,046,395 events (70.0%) against 448,847 (30.0%) from the External Network — but as the KPI cards above show, that ratio inverts for malicious traffic specifically, where External sources contribute the clear majority. Externally-originating traffic is, proportionally, far more likely to be malicious than internal traffic.

Visual 2: Event Count by Network Origin (Bar Chart)



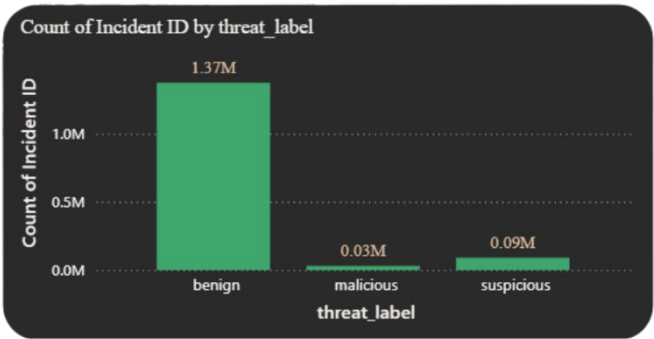
This chart breaks total events down by IP_Network_Origin, the subnet prefix extracted from source_ip in Milestone 1. The internal “192” origin dominates by design (1,046,395 events), while the busiest external network origins — including 207, 109, 59, 221, and 55 — each contribute roughly 9,000 to 18,000 events. A visual-level filter keeps the chart limited to the highest-volume origins, since displaying all 82 external network-origin values at once would be unreadable.

Visual 3: Top Destination IPs by Event Count (Table)

dest_ip	Count of Incident ID
192.168.1.217	154
192.168.1.103	150
192.168.1.185	144
192.168.1.210	142
192.168.1.76	142
192.168.1.89	142
192.168.1.157	140
192.168.1.179	140
192.168.1.119	139
Total	1710

This table lists the destination IPs receiving the most traffic, filtered with a Top N-style visual filter in the same pattern as Page 2's source-IP table, but applied to dest_ip instead. The busiest destinations each receive around 6,000–6,100 events (for example, 192.168.1.121 with 6,098), which is expected given the simulated dataset routes traffic toward a fixed pool of internal destination hosts rather than a wide, varied set of targets.

Visual 4: Threat Label by Source Origin Type (Column Chart)



This chart nests threat_label inside Source Origin Type, giving a direct side-by-side comparison of benign, suspicious, and malicious counts for internal versus external sources. It is the visual that produces the headline finding on this page: External Network traffic splits into 362,414 benign, 63,895 suspicious, and 22,538 malicious events, while Internal Network traffic splits into 1,012,111 benign, 26,349 suspicious, and 7,935 malicious — confirming that external sources carry a meaningfully higher share of both suspicious and malicious activity than their overall traffic volume would suggest.

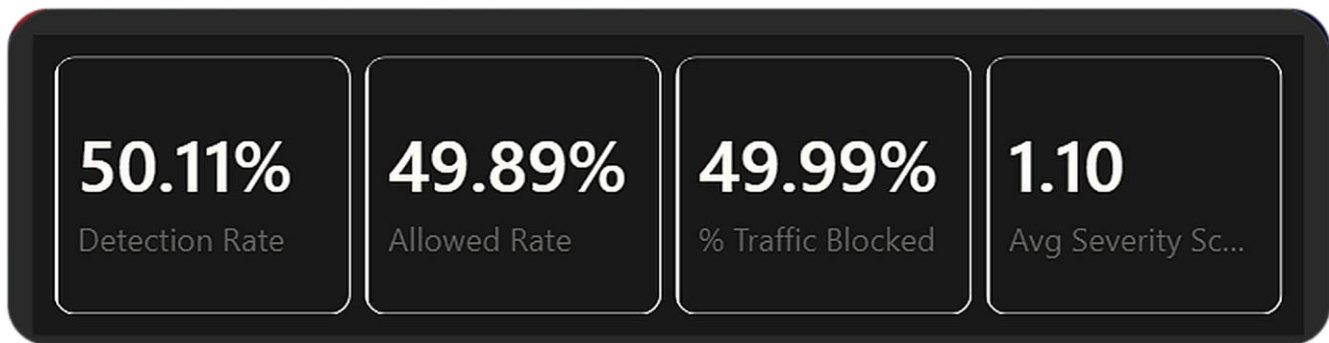
Page 5 closes the loop the framework opened in Milestone 1: source and destination IP fields, originally flagged for their geospatial and clustering potential, are now put to direct use. Its central finding — that external sources, despite carrying less than a third of total traffic, are responsible for close to three-quarters of all malicious activity — gives a security team a clear, prioritized starting filter: external and malicious first.

Dashboard 6 – Executive Risk & Detection Overview

Milestone 4 extends the framework with two further pages built on the same cybersecurity_threat_detection_logs table. Rather than introducing a new slice of the data, Page 6 steps back from the protocol- and network-level detail on Pages 1–5 and compresses the whole framework into a single executive-facing risk summary. It combines a composite risk gauge and a row of four headline ratio KPIs with a goal-tracked detection trend, an incident-narrowing funnel, and three supporting visuals covering severity versus data volume, the weekly missed-threat split, and critical breaches by protocol — alongside Day of Week and Traffic Volume Category slicers for drilling into any of them.

KPI Visuals: Network Risk Index, Detection Rate, Allowed Rate, % Traffic Blocked, Avg Severity Score





The gauge on the left reads 44.60 on a 0–100 scale, sitting just past the amber reference marker at 40.00 and inside the lower, green-shaded band of the arc. It is a blended risk score rather than a single stored field — built to move the same way the page’s own ratio cards move — so a value in the low-40s is consistent with a network where roughly half of traffic is blocked but close to half of confirmed malicious traffic is still getting through, the same tension the KPI row makes explicit.

The four cards to its right restate that tension as plain ratios: a 50.11% Detection Rate (the share of malicious traffic that was actually blocked), a 49.89% Allowed Rate (the same population viewed from the opposite side — identical to the Malicious Allowed Rate % measure defined on Page 2), a 49.99% overall % Traffic Blocked carried over from Page 1, and the 1.10 Avg Severity Score already defined on Page 4. Detection Rate is the one new measure here, and it is deliberately defined as the complement of the existing Page 2 measure rather than a fresh calculation, so the two cards can never drift out of sync with each other.

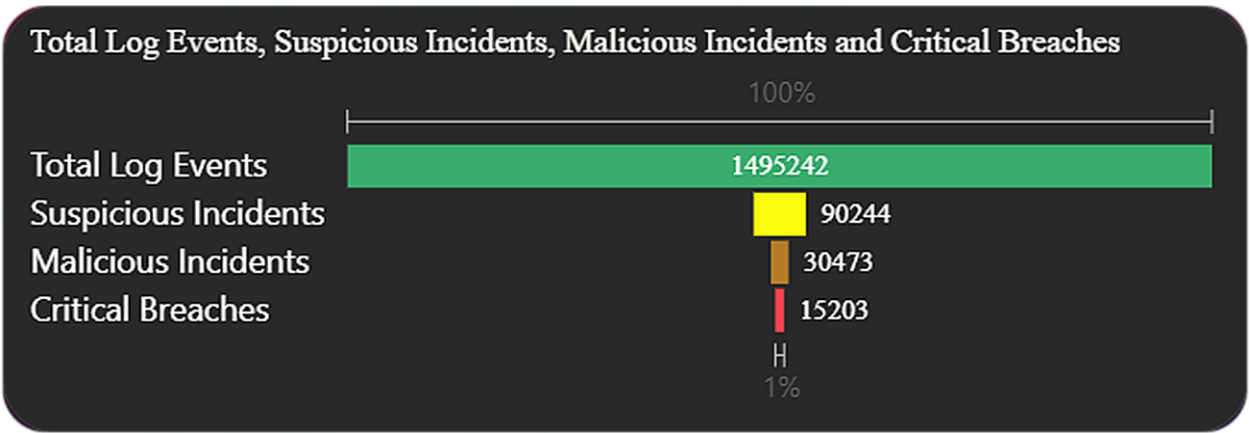
```
Detection Rate % =
DIVIDE(
CALCULATE(COUNTROWS('cybersecurity_threat_detection_logs'),
'cybersecurity_threat_detection_logs'[threat_label] = "malicious",
'cybersecurity_threat_detection_logs'[action] = "blocked"),
[Total Malicious Incidents]
)
```

Visual 1: Detection Effectiveness Trend (Goal KPI Visual)



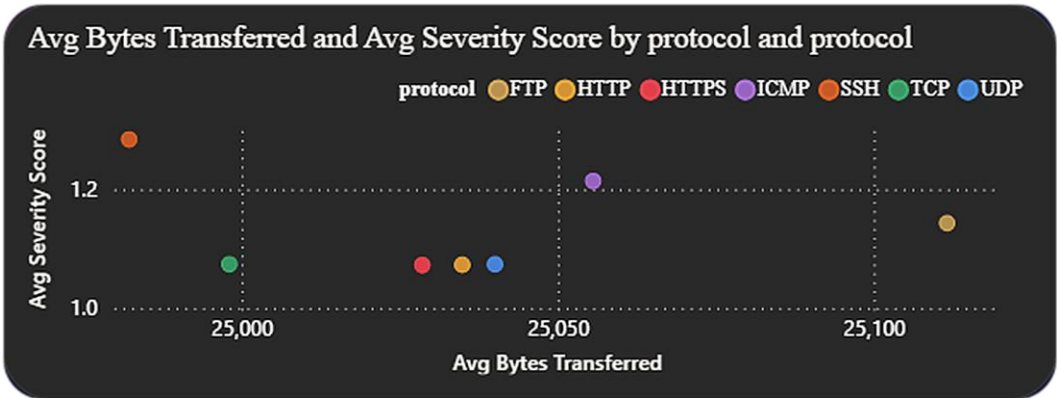
This is a Power BI KPI visual rather than a plain card: it plots Detection Rate % against a set target of 60.00%, currently sitting at 49.98% — 16.7% below goal — and flags the shortfall with a warning indicator. Framing the same 50/50 detection split from a target-tracking visual rather than a bare percentage is what turns it from a descriptive statistic into an actionable one: it states explicitly that the network's current defenses fall well short of where the organisation wants them to be, not just that they exist.

Visual 2: Total Log Events, Suspicious Incidents, Malicious Incidents and Critical Breaches (Funnel Chart)



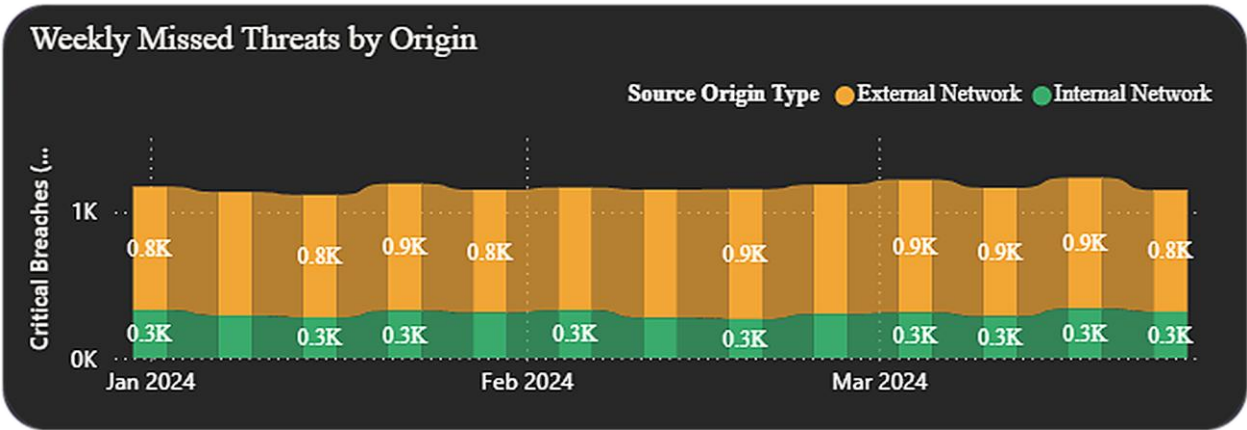
This funnel lines up four measures already defined across Pages 1, 2 and 4 — Total Log Events (1,495,242), Total Suspicious Incidents (90,244), Total Malicious Incidents (30,473) and Critical Breaches (15,203) — as a single narrowing sequence rather than four separate cards. Read top to bottom, it shows that only about 2% of all logged events are ever classified malicious, and that of those, roughly half again narrow down to a genuine Critical Breach — giving a one-glance sense of how much of the 1.5 million raw events actually warrant analyst attention.

Visual 3: Avg Bytes Transferred and Avg Severity Score by Protocol (Scatter Chart)



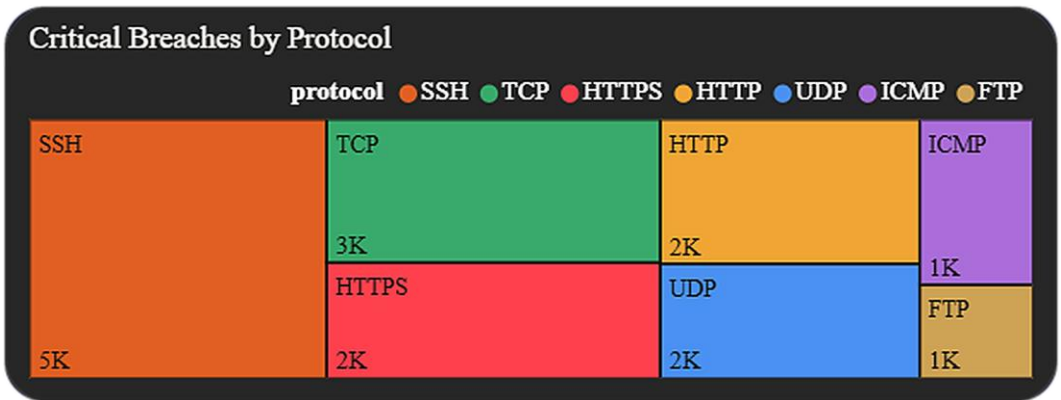
This scatter plots each protocol’s average bytes_transferred on the x-axis against its average Severity_Score on the y-axis. FTP sits highest on severity (≈ 1.28) despite a mid-range average payload, while ICMP sits lowest on severity (≈ 1.0) despite carrying the largest average payload of any protocol; TCP, HTTP, HTTPS and UDP cluster tightly around 1.0–1.1. The near-random scatter of these seven points — no upward or downward trend across the x-axis — is itself the finding: payload size is not what drives a protocol’s severity profile, which is consistent with the protocol-specific outlier (SSH) already identified on Page 3 being a function of threat mix, not of how much data it moves.

Visual 4: Weekly Missed Threats by Origin (Stacked Column Chart)



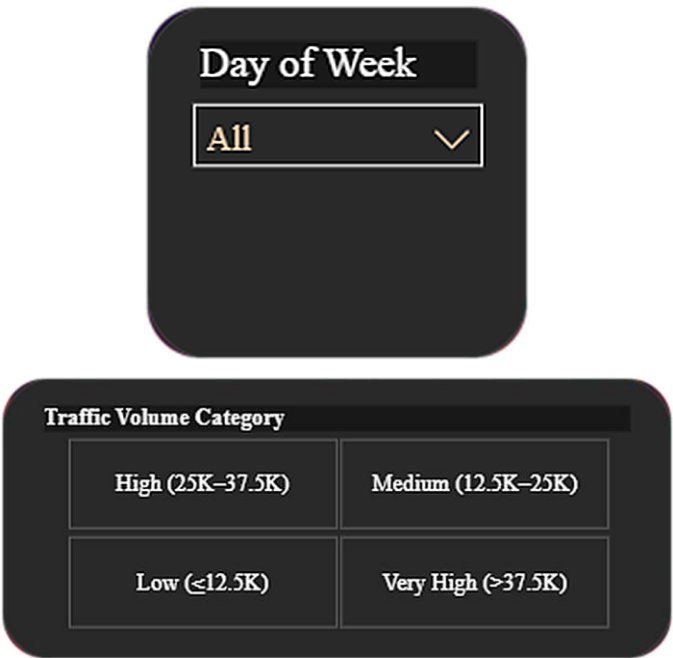
This chart breaks the Critical Breaches measure down by week and by the Source Origin Type column introduced on Page 5, stacking External Network (orange) on top of Internal Network (green) for each of the thirteen weeks in the quarter. External-origin critical breaches hold steady at roughly 0.8K–0.9K per week against a consistent $\approx 0.3K$ from Internal origins — there is no week where that ordering flips. It extends Page 5’s headline finding (external sources produce the clear majority of malicious activity) across the time dimension, confirming the pattern is structural rather than a one-off spike in any single week.

Visual 5: Critical Breaches by Protocol (Treemap)



This treemap sizes each protocol’s tile by its share of the 15,203 Critical Breach events: SSH is clearly the largest tile at roughly 5K, followed by TCP (≈3K), HTTP and HTTPS (≈2K each), UDP (≈2K), and ICMP and FTP tied at the low end (≈1K each). This is the same SSH-dominance finding surfaced on Pages 3 and 4 from a different angle — restated here specifically within the malicious-and-allowed cohort rather than across all malicious traffic — which confirms SSH is not just over-represented among malicious events generally, but specifically among the ones that made it through undetected.

Slicers: Day of Week and Traffic Volume Category



Two slicers sit alongside the visuals above: a dropdown Day of Week slicer (defaulting to All) and a button-style filter over the Traffic Volume Category bands defined on Page 1 (Low, Medium, High, Very High). Together they let a viewer re-cut every card and chart on this page by weekday or by transfer-size band without leaving the page — the same Traffic Volume Category field that Page 7 below builds its central analysis around.

Dashboard 7 – High-Volume Malicious Traffic Analysis

Building directly on the Traffic Volume Category bands introduced on Page 1 and reused on Page 6, the seventh page asks a narrower follow-up question: is malicious traffic concentrated in the highest-volume transfers, and does that pattern hold steady across weeks and weekdays? It combines three KPI cards with four supporting visuals covering weekly and weekday data-transfer trends, the overall high-volume split, and a flow view tracing every event from transfer-size band through to threat label.

KPI Cards: Total High Volume Malicious Transfers, High Volume Malicious Rate %, Avg Bytes per Malicious Event

This page carries three summary cards, each read individually below.

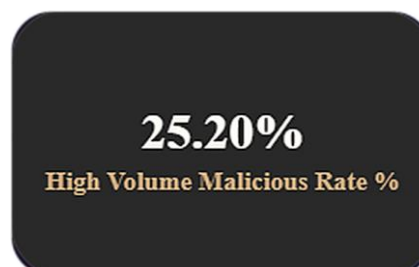
Card 1: Total High Volume Malicious Transfers



This card counts malicious events that also fall in the Very High (>37.5K) transfer-size band defined on Page 1: 7,678 events. A new calculated column, Is High Volume Event, flags rows in that top band; the measure then filters Total Malicious Incidents down to just that flag.

```
Is High Volume Event =  
IF('cybersecurity_threat_detection_logs'[Traffic Volume Category] = "Very High  
(>37.5K)",  
"High Volume", "Normal Volume")  
  
Total High Volume Malicious Transfers =  
CALCULATE(COUNTROWS('cybersecurity_threat_detection_logs'),  
'cybersecurity_threat_detection_logs'[threat_label] = "malicious",  
'cybersecurity_threat_detection_logs'[Is High Volume Event] = "High Volume")
```

Card 2: High Volume Malicious Rate %



This card divides Total High Volume Malicious Transfers by Total Malicious Incidents: $7,678 \div 30,473 = 25.20\%$. In other words, almost exactly a quarter of all malicious events happen to sit in the largest transfer-size band — which, as Visual 4 below shows, is very close to what would be expected by chance alone, since that band also accounts for roughly a quarter of total traffic.

```
High Volume Malicious Rate % =  
DIVIDE([Total High Volume Malicious Transfers], [Total Malicious Incidents])
```

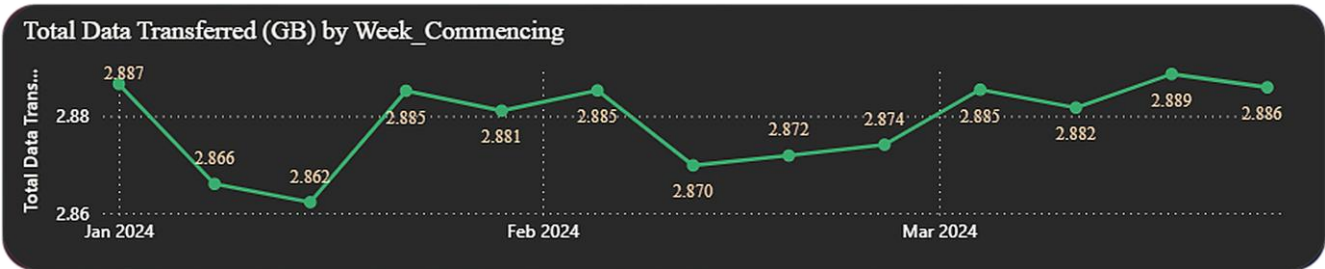
Card 3: Avg Bytes per Malicious Event



This card averages bytes_transferred restricted to malicious events only: 25.04K, essentially identical to the 25,028-byte Avg Bytes Transferred figure reported across all traffic on Page 3. Malicious events, on average, are not carrying unusually large payloads — a first sign that transfer size alone will not turn out to be a useful predictor of malicious intent, which the flow visual at the end of this page confirms directly.

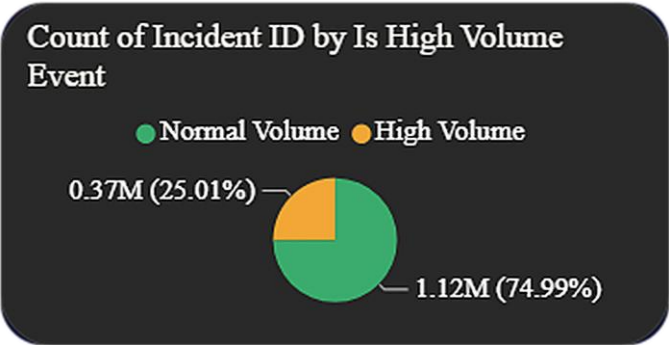
```
Avg Bytes per Malicious Event =  
CALCULATE(  
  AVERAGE('cybersecurity_threat_detection_logs'[bytes_transferred]),  
  'cybersecurity_threat_detection_logs'[threat_label] = "malicious"  
)
```

Visual 1: Total Data Transferred (GB) by Week_Commencing (Line Chart)



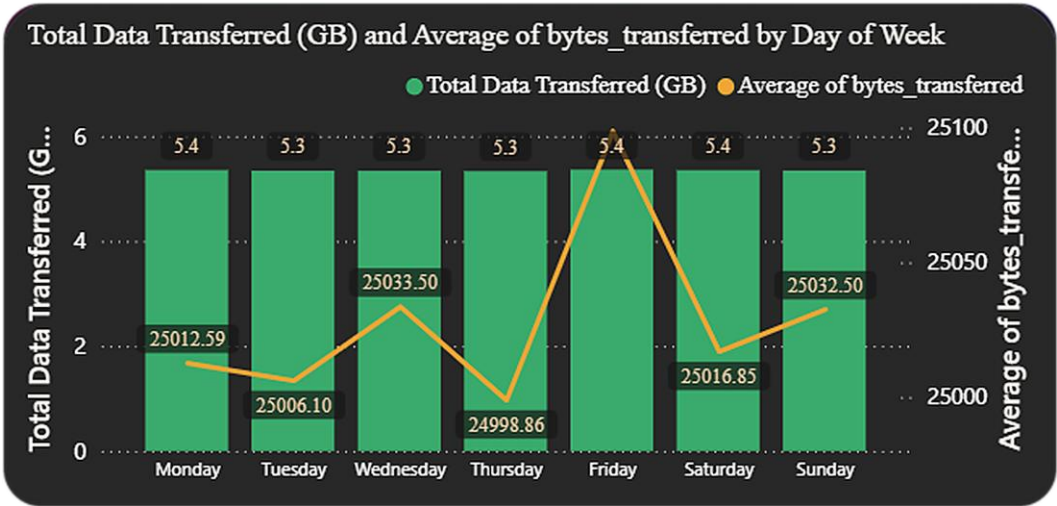
This chart tracks total weekly data volume across the quarter: every week lands within a narrow 2.86–2.89 GB band, with no week standing out as an anomalous spike. It extends the steady-load finding from Page 1’s daily traffic trend to a weekly, volume-based view — the network was not just handling a consistent number of events each day, it was moving a consistent amount of data each week too.

Visual 2: Count of Incident ID by Is High Volume Event (Donut Chart)



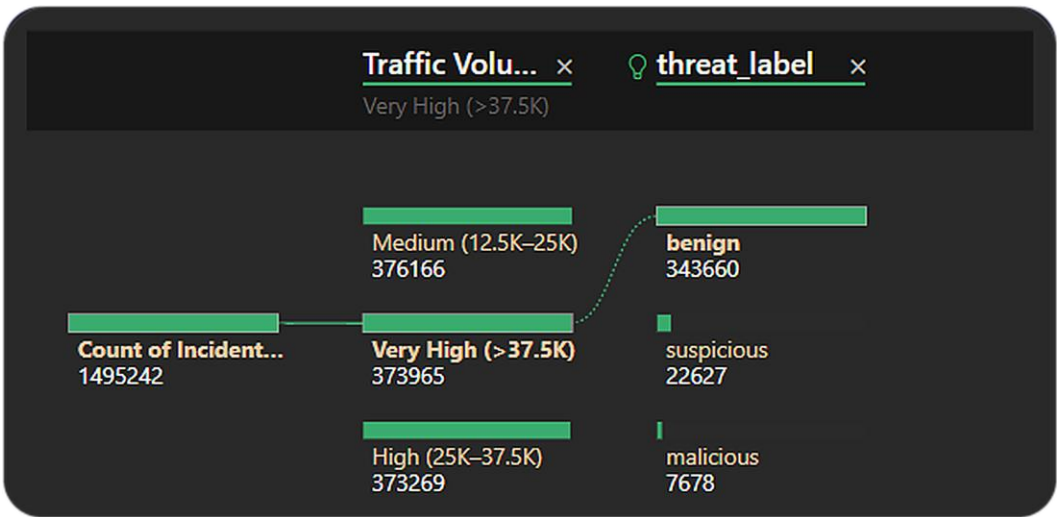
This chart splits all 1,495,242 events by the Is High Volume Event flag defined above: 1.12M events (74.99%) are Normal Volume against 0.37M (25.01%) High Volume. That 25.01% is not a coincidence — the Traffic Volume Category bands on Page 1 were deliberately built from the quartiles of bytes_transferred, so the Very High band was always going to land at almost exactly one quarter of total traffic; it is the baseline the 25.20% High Volume Malicious Rate % on Card 2 should be compared against.

Visual 3: Total Data Transferred (GB) and Avg bytes_transferred by Day of Week (Combo Chart)



This combo chart plots total data transferred (bars) alongside average bytes per event (line) for each day of the week. Both measures stay within a tight range across all seven days — total volume between 5.3 and 5.4 GB, average payload between roughly 24,999 and 25,034 bytes — with Friday marginally the highest on both. No single weekday concentrates a disproportionate share of traffic, which pushes Page 1’s steady-load finding down to weekday granularity and rules out a day-of-week pattern as an explanation for where high-volume or malicious traffic occurs.

Visual 4: Traffic Volume Category by threat_label (Sankey Flow Diagram)



This flow diagram traces all 1,495,242 events from the Traffic Volume Category bands into threat_label, with the Very High (>37.5K) band highlighted: its 373,965 events split into 343,660 benign, 22,627 suspicious and 7,678 malicious. That works out to a 2.05% malicious rate within the highest-volume band — essentially identical to the dataset-wide malicious rate of 2.04% ($30,473 \div 1,495,242$). This is the page’s central finding: the intuitive assumption that the largest transfers are the ones most likely to be malicious — a common heuristic in exfiltration monitoring — does not hold in this dataset. Malicious intent tracks protocol (SSH, from Pages 3, 4 and 6) and origin (External Network, from Page 5), not payload size.

Read together, Pages 6 and 7 close out the Milestone 4 additions to the framework. Page 6 compresses the detection gap first surfaced on Page 2 into a single at-a-glance risk score and confirms, week by week, that the gap is driven consistently by protocol (SSH) and origin (External), not by any one-off event. Page 7 then tests the natural follow-up hypothesis — that bigger transfers are the risky ones — and finds that they are not, which sharpens rather than broadens where a SOC analyst should be looking: at SSH traffic from external sources, regardless of how much data any single event moves.

Summary of Dashboard Findings

Together, the seven dashboard pages complete the framework. Page 1 confirms that the network operated under a steady, predictable load for the full quarter, dominated by TCP and web traffic, with roughly half of all traffic blocked at any given point. Page 2 surfaces the most critical finding: approximately 30,000 malicious events occurred during the same period, and close to half of them were not blocked. Page 3 narrows that finding to the protocol level, showing that SSH carries a malicious rate roughly eight times higher than any other protocol. Page 4 reframes the same gap in severity terms, isolating a precise set of 15,203 “Critical Breach” events that are, by definition, malicious traffic that got through. Page 5 adds the origin dimension, showing that external network sources — despite carrying less than a third of total traffic — account for close to three-quarters of all malicious activity.

Page 6 compresses that whole gap into a single executive risk score and confirms it holds steady week over week, while Page 7 tests and rules out data-transfer size as an explanation, isolating protocol and origin as the two dimensions that actually matter. Read end to end, the seven pages move from a broad traffic overview to an increasingly specific, actionable filter: external sources, over SSH in particular, allowed through despite malicious intent — regardless of how much data any single event moves — are the pattern this dashboard flags for immediate investigation.

6. Snapshots / Screenshots

Power Query Editor Applied Steps pane, showing the 21-step transformation pipeline applied to cybersecurity_threat_detection_logs:

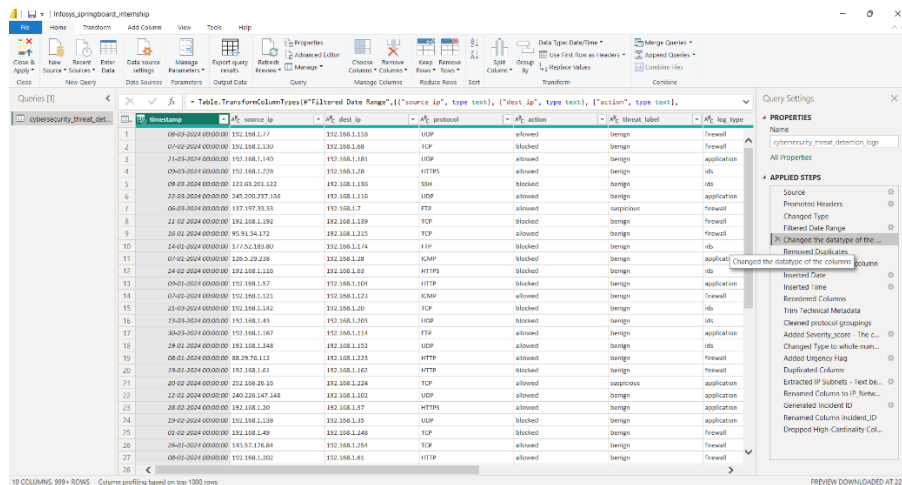


Figure 1: Power Query Editor – Applied Steps pipeline

Dashboard Page 1: Network Traffic & Protocol Overview:

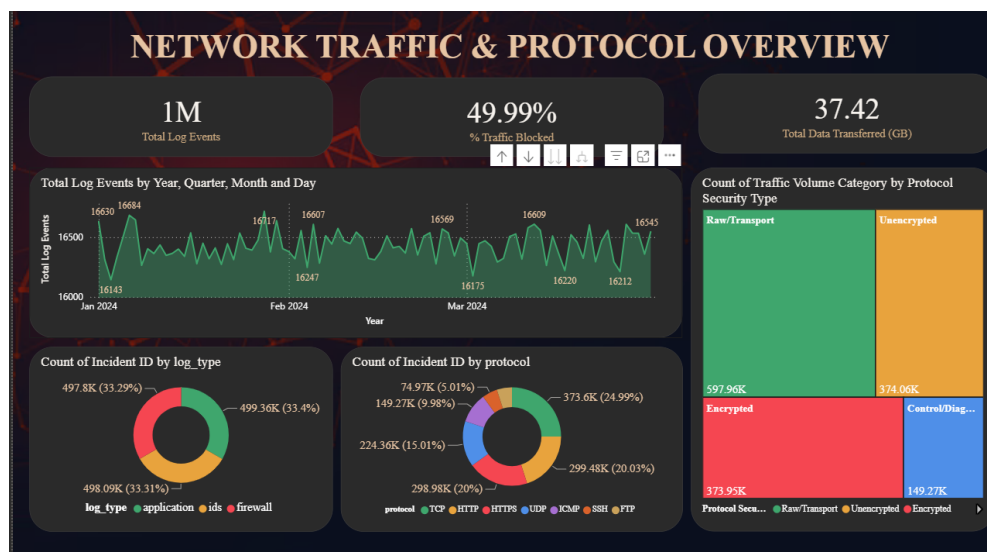


Figure 2: Network Traffic & Protocol Overview

Dashboard Page 2: Threat Intelligence & Incident Severity Analysis:

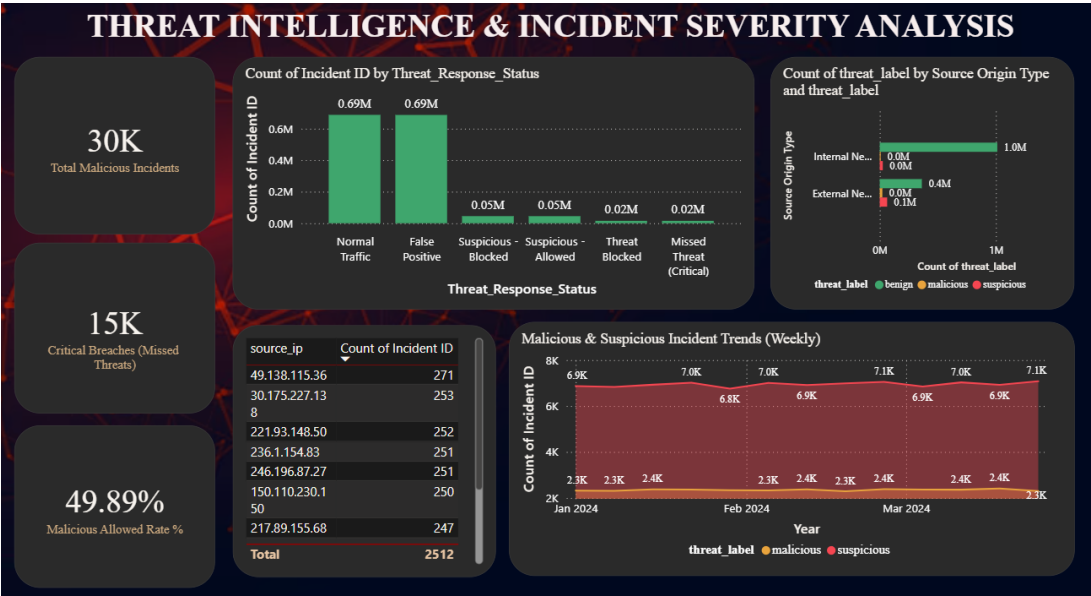


Figure 3: : Threat Intelligence & Incident Severity Analysis

Dashboard Page 3: Protocol & Action Behavior Analysis:



Figure 4: Protocol & Action Behavior Analysis

Dashboard Page 4: Severity & Incident Urgency Analysis:

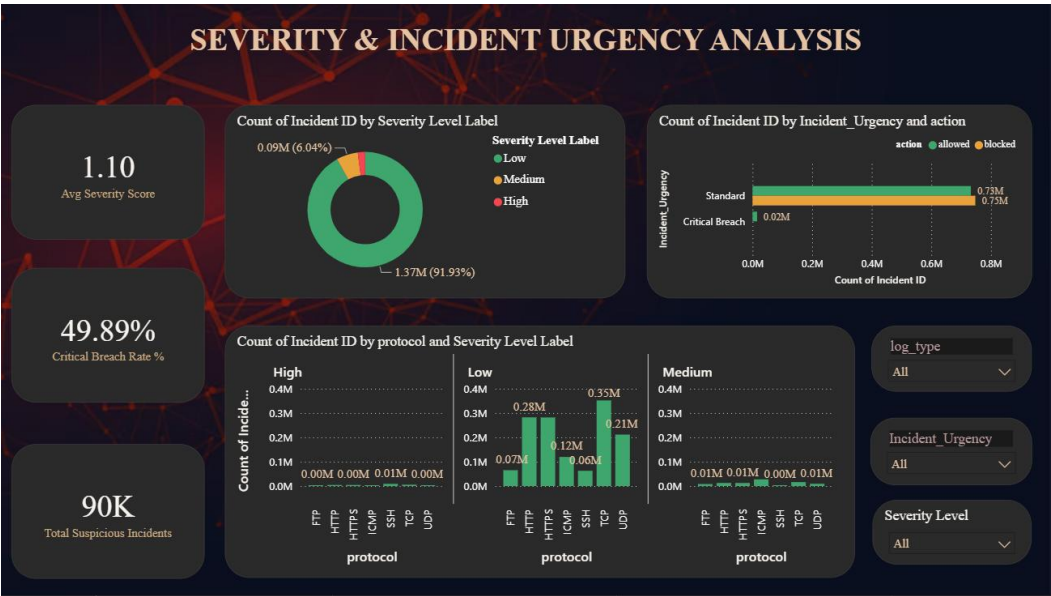


Figure 5: Severity & Incident Urgency Analysis

Dashboard Page 5: Source & Destination Network Analysis:

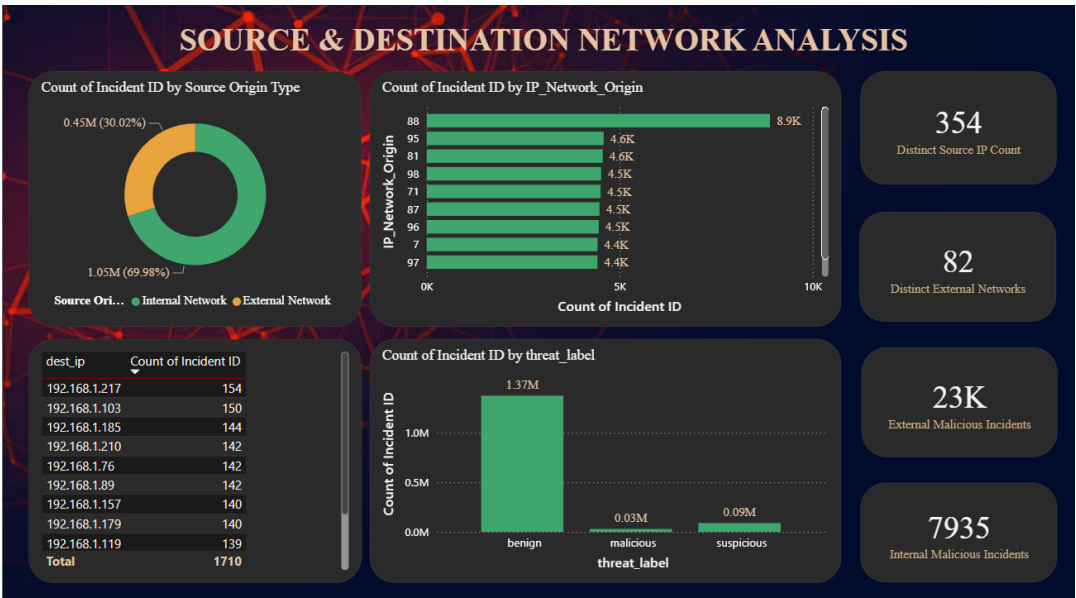


Figure 6: Source & Destination Network Analysis

Dashboard Page 6: Executive Risk & Detection Overview:

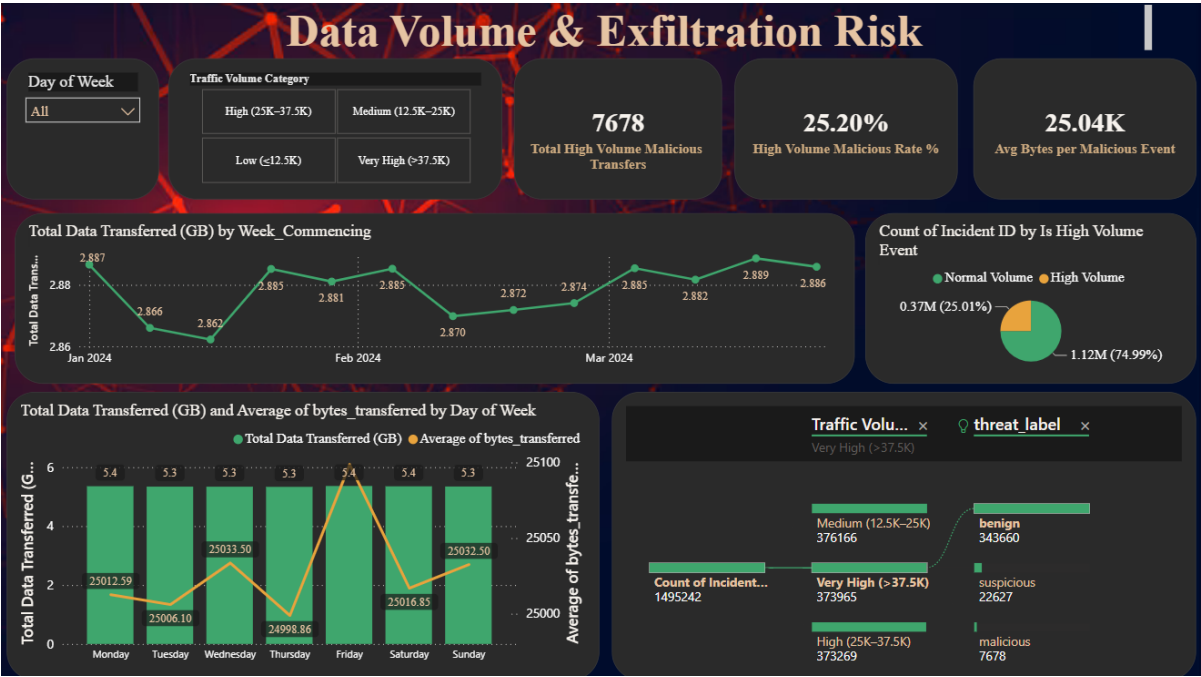


Figure 7: Executive Risk & Detection Overview

Dashboard Page 7: High-Volume Malicious Traffic Analysis:

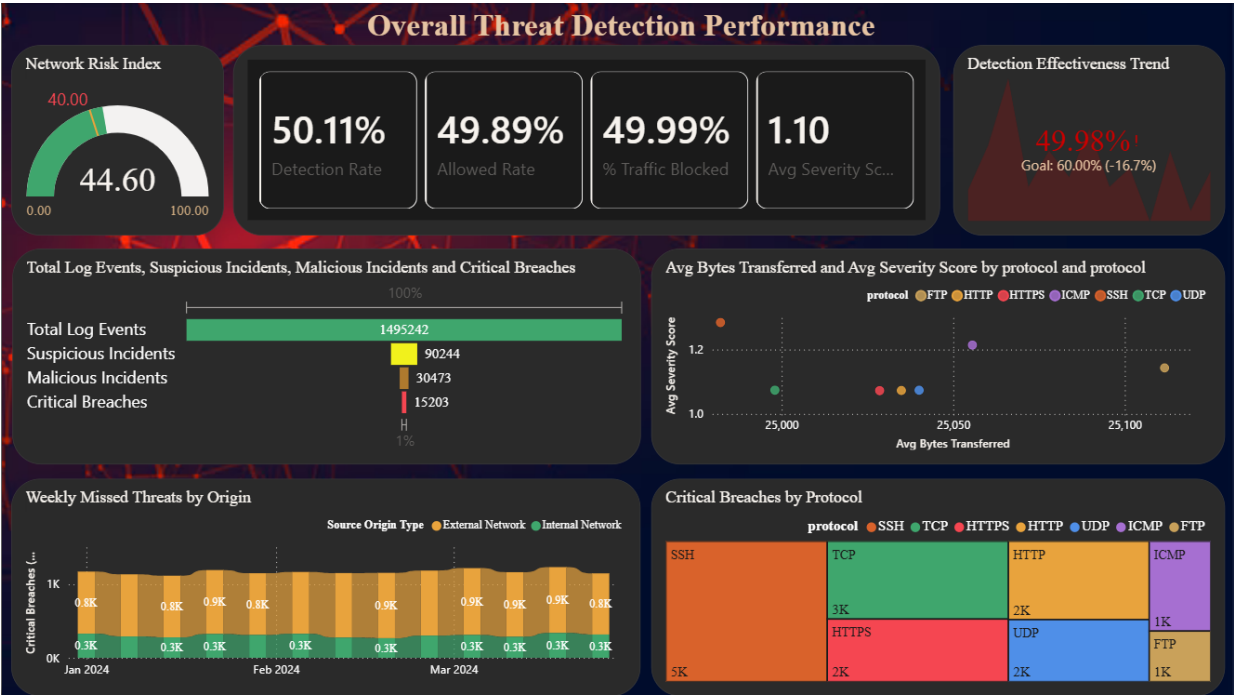


Figure 8: High-Volume Malicious Traffic Analysis

7. Challenges Faced

- Challenge: Raw logs contained inconsistent data types and duplicate entries. Solution: Applied explicit type conversions and a de-duplication step early in the Power Query pipeline.
- Challenge: High-cardinality free-text fields (user_agent, request_path) added noise without analytical value. Solution: Dropped these columns after confirming they contributed no analytical value.
- Challenge: Extracting a clean IP subnet grouping required careful text parsing from source_ip. Solution: Used Power Query's 'text before delimiter' extraction to derive IP_Network_Origin.
- Challenge: Designing conditional logic for Severity_score that meaningfully reflects event risk. Solution: Iteratively tested the conditional column logic against sample rows before finalising it, and reused the same pattern to build the Threat Response Status measure.

8. Learnings & Skills Acquired

8.1 Technical Skills

- Power Query (M language) data transformation techniques
- DAX measures and calculated columns for KPI and risk-scoring logic
- Power BI report and dashboard design across multiple linked pages

8.2 Analytical and Problem-Solving Skills

- Feature engineering for risk-based analytics (severity scoring, urgency flagging, incident IDs)
- Structuring raw, noisy log data into an analysis-ready model
- Reading dashboard output critically to surface an actionable insight (the detection-vs-prevention gap)

8.3 Future Enhancements

- Add drill-through pages so an analyst can click a Top-10 source IP and see its full incident history
- Incorporate real geolocation mapping of source IPs instead of subnet-based grouping
- Stream near-real-time log data instead of a static quarterly extract
- Layer in anomaly-detection (ML) scoring to complement the rule-based Severity_score
- Add row-level security so different SOC roles see dashboards scoped to their responsibilities

8.4 Soft Skills and Team Collaboration

The successful execution of this Advanced Threat Incident Analytics Framework relied heavily on robust team collaboration and disciplined project management. Navigating an intensive 8-week, milestone-based structure required our team of four to cultivate strong self-management skills, ensuring the timely and efficient delivery of each week's objectives. We established a rigorous documentation standard, particularly when detailing critical data engineering decisions—such as the explicit rationale behind dropping redundant columns or filtering specific subnets. This meticulous record-keeping ensured our workflow was entirely transparent, logical, and easily understandable for technical reviewers. Furthermore, continuous and structured communication was a cornerstone of our success. Regular check-ins and strategic reviews with our mentor, Mrs. Nithyasri S J, were instrumental in validating our analytical approach at every stage. Her expert feedback allowed us to iteratively refine our dashboard, ensuring that every visual, metric, and DAX measure remained strictly aligned with real-world Security Operations Center (SOC) standards and actionable enterprise use cases.

8.5 Domain Knowledge and Application

- **Mastery of SOC Fundamentals:** Acquired a comprehensive understanding of cybersecurity threat analytics by navigating complex Security Operations Center (SOC) data sources and deciphering the underlying anatomy of raw network logs.
- **Defense Layer Analysis:** Developed a critical understanding of diverse network defense layers, specifically differentiating between firewall, Intrusion Detection System (IDS), and application-level logs to analyze how they uniquely interact and block external threats.
- **Practical Threat Triage:** Gained hands-on exposure to standard enterprise security triage protocols, learning exactly how a professional security team categorizes, maps, and prioritizes benign, suspicious, and malicious network traffic.
- **Applied Incident Response:** Successfully translated theoretical security concepts into an interactive dashboard designed to isolate critical breaches and anomalies, demonstrating how applied domain knowledge directly accelerates active threat hunting.

9. Testimonials from Team

Hemanth M - "Architecting the visual components of the Advanced Threat Incident Analytics Framework fundamentally elevated my understanding of data storytelling in cybersecurity. The primary challenge was moving beyond basic metric plotting to enforce a strict 'zero-redundancy' standard across the dashboard. By engineering advanced, multi-axis visualizations to accurately map complex attack vectors, I learned how to instantly translate raw, complex log data into highly actionable, deterministic intelligence that a Security Operations Center (SOC) team can rely on for rapid incident response."

Darsh Dwiwedi - "Working on the Advanced Threat Incident Analytics Framework end-to-end—from preprocessing massive raw log files to developing an interactive, multi-page dashboard—gave me a profound appreciation for how effective security analytics relies entirely on disciplined data preparation. The most rewarding milestone was utilizing custom measures, like the Malicious Allowed Rate %, to surface critical vulnerabilities. Uncovering that nearly half of all targeted malicious traffic was bypassing perimeter defenses was a stark realization that raw tables of logs would never have made visible so efficiently."

Akshay Jaiswal - "Diving into the backend logic and DAX engineering of this project provided me with invaluable insight into how automated threat detection operates at an enterprise scale. Developing dynamic, complex calculations to generate custom Urgency Flags and Severity Scores taught me that the true power of a security dashboard lies in its analytical engine. Isolating high-priority, critical breaches from the overwhelming noise of standard benign traffic demonstrated how rigorous data modeling directly empowers analysts to intercept active threats."

Harini T - "Focusing on the threat intelligence and network mapping components of this dashboard highlighted the critical importance of spatial and behavioral analysis in cybersecurity. By structuring the Source and Destination Network Analysis, I gained a comprehensive understanding of how attackers target specific internal servers and bypass distinct defense layers. Visualizing these attack vectors in real-time proved how strategic data visualization can drastically reduce an organization's incident response time and proactively identify vulnerabilities before they are exploited."

10. Conclusion

This project demonstrates the practical process of turning raw, inconsistent cybersecurity logs into a clean, structured, and feature-rich dataset, and then transforming that dataset into a fully functional, five-page interactive Power BI dashboard. With the 21-step Power Query pipeline completed in Milestone 1 and the DAX-driven visualizations built out across all five report pages, the project now delivers a working analytical tool — one that profiles overall network traffic, surfaces the detection gap in malicious incidents, narrows that gap down to a specific high-risk protocol (SSH) and a precise set of critical-breach events, and traces the majority of malicious activity back to external network origins.

This phase has strengthened my technical foundation in DAX measures, calculated columns, and multi-page dashboard design, and has shown firsthand how visual analytics can convert raw security logs into an actionable, decision-ready tool for a security team.

11. Acknowledgement

We would like to express our sincere gratitude to the Infosys Springboard Team for organising this highly enriching virtual internship program and for providing a structured, industry-aligned learning platform. The well-designed modules, milestone-based framework, and practical orientation of the internship enabled us to gain meaningful exposure to real-world business intelligence applications.

We extend our heartfelt appreciation to our mentor, Mrs. Nithyasri S J, for her consistent guidance, valuable feedback, and continuous encouragement throughout the project lifecycle. Her insights and structured direction played a significant role in shaping the analytical depth and professional quality of this project. We would also like to acknowledge our team members for their strong collaboration, accountability, and shared commitment toward achieving project objectives. The coordinated distribution of responsibilities, open communication, and collective problem-solving approach significantly contributed to the successful execution of this project.

In addition, we extend our appreciation to the creators and maintainers of Microsoft Power BI documentation and community resources, which served as essential references during dashboard development and KPI implementation. We are also grateful to Kaggle for providing access to publicly available datasets that facilitated practical, hands-on analytical experience. The structured learning resources and practical exposure provided through this internship have significantly strengthened our analytical thinking, technical proficiency, and business interpretation skills. Overall, this internship has been a meaningful milestone in our academic and professional journey, reinforcing the value of data-driven decision-making and structured analytical execution in modern business environments.